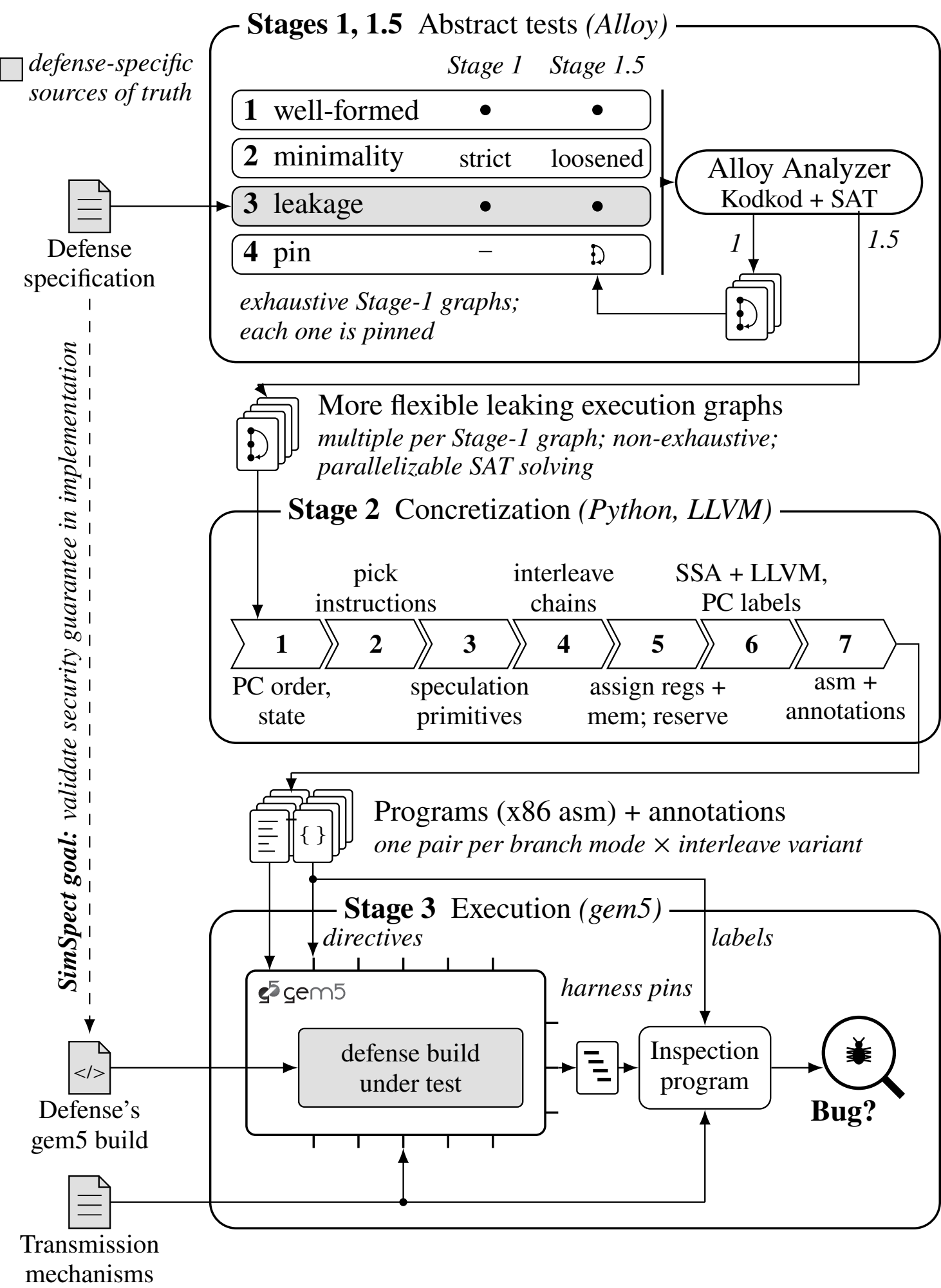

defense-specific sources of truth
Stages 1, 1.5 Abstract tests (Alloy)
Stage 1
Stage 1.5
1 well-formed
2 minimality
strict
loosened
3 leakage
4 pin
exhaustive Stage-1 graphs; each one is pinned
Alloy Analyzer
Kodkod + SAT
1
1.5
Defense specification
SimSpect goal: validate security guarantee in implementation
More flexible leaking execution graphs
multiple per Stage-1 graph; non-exhaustive; parallelizable SAT solving
Stage 2 Concretization (Python, LLVM)
pick instructions
interleave chains
SSA + LLVM, PC labels
1
2
3
4
5
6
7
PC order, state
speculation primitives
assign regs + mem; reserve
asm + annotations
Programs (x86 asm) + annotations
one pair per branch mode × interleave variant
Stage 3 Execution (gem5)
directives
labels
harness pins
gem5
defense build under test
Inspection program
Bug?
Defense's gem5 build
Transmission mechanisms